

Ambulance Dispatch Delay Risk Prediction

AIML project for emergency response decision support

- Goal: predict elevated ambulance dispatch or arrival delay risk.
- Output: probability, risk band, binary flag, and operational recommendations.
- Prototype includes data pipeline, ML models, evaluation, and Streamlit app.

Problem and Impact

Why this matters

- Ambulance delays can affect emergency outcomes and public-safety operations.
- Dispatchers need early warning when a case is delay-prone.
- Managers can use risk patterns to improve zone coverage and resource planning.
- The system supports decisions; it does not replace dispatcher judgment.

Dataset and Inputs

IERAD-compatible emergency response data

- Reference dataset: Kaggle Integrated Emergency Response Dataset (IERAD).
- Repository includes normalized data/emergency_response_data.csv.
- Synthetic fallback generator is included for environments without Kaggle access.
- Inputs include priority, incident type, time, weather, traffic, zone, distance, crew, vehicle, temperature, zone delay rate, and caller stress.

System Workflow

From raw data to user action

- 1. Load IERAD-compatible or synthetic records.
- 2. Normalize fields into the project schema.
- 3. Validate data and split into train, validation, and test sets.
- 4. Fit preprocessing: scaling, one-hot encoding, and ordinal encoding.
- 5. Train, evaluate, save models, and serve predictions in Streamlit.

AI / ML Component

Delay-risk classification

- Logistic regression baseline with one-hot categorical features.
- Fully connected neural network with integer-encoded categorical features.
- Fully connected neural network with embedding layers for categorical features.
- Validation threshold tuning converts probabilities into practical risk flags.
- ML is useful because delay risk depends on interacting operational factors.

Prototype and User Flow

Streamlit decision-support app

- Overview page shows dataset and delay-risk summaries.
- Model dashboard shows metrics, thresholds, plots, and feature importance.
- Single prediction form accepts one emergency case and returns risk output.
- Batch prediction supports CSV upload and downloadable prediction results.
- About page documents intended use and responsible-use cautions.

Results

Saved model comparison

- Logistic Regression: 97.57% accuracy, 99.79% AUC-ROC.
- FCNN Integer Encoding: 98.36% accuracy, 99.91% AUC-ROC.
- FCNN Embedding Layers: 98.66% accuracy, 99.95% AUC-ROC.
- Saved figures include confusion matrices, ROC curves, metric comparison, and training history.
- Best saved model: FCNN with embedding layers.

Sample Output and Actions

What the user receives

- Delay probability and risk band: low, moderate, or high.
- Binary delay-risk prediction based on tuned decision threshold.
- Prediction drivers compare the case against reference values.
- Recommendations may include backup unit alert, route review, active monitoring, weather warning, or pre-arrival instructions.
- Output is written in simple operational language for decision-makers.

Limitations and Responsible Use

Safety and interpretation

- IERAD may not provide a direct binary delay-risk label, so the target can be engineered.
- Metrics demonstrate workflow quality, not proven real-world dispatch performance.
- No live GPS, traffic feeds, ambulance availability, or hospital operations integration yet.
- Predictions must be validated on local real dispatch data before operational use.
- Human dispatcher judgment and emergency service protocols remain primary.

Future Improvements and Conclusion

Next steps

- Use true response-time or arrival-delay labels when available.
- Add response-time regression in minutes and geospatial delay hotspot maps.
- Integrate live traffic, GPS, unit availability, and hospital capacity feeds.
- Add stronger interpretability such as SHAP and more automated tests.
- Conclusion: the repo delivers a working AIML prototype with data, models, evaluation, app, recommendations, and documentation.